

Proxmox VE: The One Stop Open Source Platform for Virtualisation

Proxmox VE is a simple yet powerful all-in-one solution with all the major features needed for any IT setup to manage virtual infrastructure. It not only comes with support for backup/restore or replication/HA but also the Hyper Converge Infra (HCI) setup for storage with multi-master clusters. Being open source, Proxmox is a solution with low administrative and support costs.



Systems admins face numerous challenges today when it comes to managing IT infrastructure and data. Virtualisation software is helping them to address these by making available virtual computer hardware platforms, virtual storage devices and virtual computer network resources. Virtualisation helps to make use of today's high-powered hardware to its fullest.

What is Proxmox VE?

Proxmox Virtual Environment is an open source platform for all-inclusive enterprise virtualisation. One can easily run Windows virtual machines (VM), Linux VMs as well as different Linux containers on it. It supports the management of software-defined storage like Ceph; does backups/restores, live migration and sync/async replication; and sets up high-

availability clustering, firewalls and disaster recovery—all this with the built-in Web management interface. It is a Debian based Linux distribution with a modified Ubuntu LTS kernel, and allows the deployment and management of virtual machines and containers.

Proxmox VE has already become the first choice for thousands of satisfied users across the globe. It serves as the best open source alternative for popular tools such as VMware vSphere, Microsoft Hyper-V or Citrix XenServer.

What makes Proxmox the right choice for enterprise virtualisation

Many companies choose an open source virtualisation solution to simplify their IT infrastructure with server virtualisation and consolidation. It lets them easily reduce operating costs as well as maintenance and administration

time. The open source platform Proxmox VE comes with zero licence costs, provides full access to all functionalities, and increases the flexibility, security and reliability of your IT infrastructure. All this makes it one of the best alternatives to proprietary enterprise solutions.

A short history

When two Linux developers, Dietmar and Martin Maurer, discovered that there was no open source GUI solution for container-virtualisation (with only OpenVZ somewhat suitable to their needs), they started developing Proxmox VE. Later they tested KVM, and realised that this technology was a more suitable hypervisor, so they started integrating the two.

The first public release took place in April 2008, and the platform quickly gained traction. It was one of the few platforms providing out-of-the-box support for container and full virtualisation, managed with a Web GUI similar to commercial offerings. Since its first release, it has evolved a lot and the current version is 5.2.

What's new in Proxmox VE 5.2

The latest release of Proxmox VE as of writing this article comes with new features like the cloud-init package for automating VM provisioning, a CIFS/SMB storage plug-in, and the *Let's Encrypt* certificate management via the graphical user interface.

The Proxmox development team has further expanded the cluster functionality, making the creation and configuration of Proxmox VE clusters possible via the Web interface. The